Activity 6 Report: Database Backup, Restoration, and Recovery

COMP 018 Database Administration | Project tested: BrewPOS (Cafe Point-of-Sale) (Spring Boot + PostgreSQL) | Date of testing: 6 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	BrewPOS (Cafe Point-of-Sale)
Owning group	Group 4: Olivarez, Charles Kean E.; Infantado, Gerwin G.; Pocsidio, Jhon Louie F.; Abangan, SJ Kaizen M.; Bartolata, Redge Raindel N.
Repository	https://github.com/Nycri/brewpos
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a6_backup.txt, a6_restore.txt, a6_verify.txt, a6_access.txt, a6_recovery.txt

This is a full re-test. The database was dropped and rebuilt from the repository files, and every result below comes from this run.

Run at 00:03 on 7 November 2026 (Asia/Manila). The original database and the dump were not modified by the restore and recovery tests.

2. Backup plan

Item	Plan
Objects	Whole database grp_brewposdb: 5 tables, 4 sequences, 9 indexes, 4 foreign keys, and the grants on tables and sequences
Method	pg_dump custom format (-Fc), the same command as database/README.md
Account	The project defines no backup role, so the dump was taken as postgres. Recommend a role with pg_read_all_data
Files outside the database	The .env file (database URL and password) is not in the repository and must be saved separately. The app stores no uploaded files
Schedule (real deployment)	Daily full dump at closing time, plus one before any schema change
Retention	7 daily and 4 weekly copies, one of them on another machine
RPO and RTO	RPO 24 hours (a day of sales could be lost), RTO 1 hour
Roles	Roles are not in a database dump. Keep pg_dumpall --roles-only next to each backup
Sensitive content	Only staff names and sales. No customer data or passwords, so the dump is low risk, but sales figures should still be kept private

3. Backup creation

```
pg_dump -U postgres -h localhost -Fc -d grp_brewposdb -f grp_brewposdb_activity6.dump
```

Item	Value
File	grp_brewposdb_activity6.dump
Size	19,076 bytes
SHA-256	852F2971705598958506ACB08BFE386B45A40575F6C917B60F5D5F67EEEC0DE3
Archive	custom format, gzip, 55 TOC entries, 5 TABLE DATA entries, table and sequence ACL entries included, dumped by pg_dump 18.6
Errors	None, exit code 0

4. Restoration and verification

```
createdb -U postgres -h localhost grp_brewposdb_restore
pg_restore -U postgres -h localhost -d grp_brewposdb_restore grp_brewposdb_activity6.dump
```

Restore succeeded with exit code 0 in 0.17 seconds and no messages. The README's own restore command (pg_restore --clean --if-exists) was then run into the same copy and also finished with no messages.

Check	Original	Restored	Match
Tables, sequences, indexes	5, 4, 9	5, 4, 9	Yes
Row counts (menu_category, menu_item, order_line_item, order_ticket, staff)	5, 15, 16, 11, 2	same	Yes
Constraints: check, foreign key, not null, primary key, unique	5, 4, 18, 5, 2	same	Yes
Table grants per role (app_admin_grp, app_readonly_grp, app_runtime_grp, brewpos_app, postgres)	17, 5, 13, 13, 35	same	Yes
md5 checksum of order_line_item	d222bce1…8c02	d222bce1…8c02	Yes
Sequence positions	19, 38, 18, 3	19, 38, 18, 3	Yes
Check constraints still enforced (quantity 0, price -1)	rejected	rejected	Yes
Application started against the restored copy	n/a	Started; pg_stat_activity shows 10 connections to grp_brewposdb_restore; GET /api/orders returned 11 orders, the same as the original	Yes

5. Access control after restore

Item	Original	Restored copy
Table and sequence privileges for brewpos_app and the group roles	present	present, same counts (they are stored in the dump)
Database-level ACL (CONNECT grants)	brewpos_app, app_runtime_grp, app_readonly_grp, app_admin_grp	empty (not restored)
brewpos_app statement_timeout and	30s and public	0 (no timeout) and "$user", public

search_path		
brewpos_app can read order_ticket / cannot delete from it	yes / yes	yes / yes
Object owner	postgres	postgres

The roles themselves live in the cluster, so on this server they were still there. On a new server they would have to be created first, or the restore prints errors for the grants. The two items that did not come back are the database-level grants and the per-role session defaults from 05_create_app_role.sql (statement_timeout 30s, lock_timeout 10s, search_path). After a restore on a new server these must be re-applied by running that script again. A diagnostic query on pg_db_role_setting failed because it used a type that does not exist in PostgreSQL 18, so the settings were checked by logging in as brewpos_app instead.

6. Recovery simulation

Step	Action	Result
Before	Checksum of order_line_item in the restored copy	16 rows, md5 d222bce1…
Incident	DELETE of the lines of the newest order directly in the database	DELETE 1, 15 rows, md5 18407772…; net sales dropped by 75.50 (the order is left with no lines)
Recovery	createdb grp_brewposdb_recovery and pg_restore of the same dump	Success, exit code 0
Compare	Checksum in the recovery database	16 rows, md5 d222bce1…, same as before the incident; net sales back to the original figure
Original untouched	Checksum in grp_brewposdb	16 rows, same md5

Outcome: recovered. A full restore into a new database was used because it leaves the damaged copy untouched for comparison. Anything sold after the last backup would be lost, which is the RPO. The net sales total in this test database (162 billion) is inflated by the quantity test of Activity 3, not by the restore.

7. Evaluation and recommendations

- The README backup and restore commands are correct and need no changes. The database is tiny, so restore takes a fraction of a second.
- Add pg_dumpall --roles-only and the 05 script to the backup procedure, since roles, database grants and session defaults are not restored from the dump.
- A daily dump is enough for a small cafe, but a shift of sales is what the cafe loses. If that is too much, enable WAL archiving for point-in-time recovery.
- Store the dump and the .env on a different machine and test a restore regularly.
- Because order history cannot be edited through the API, the most likely loss is a deletion by direct SQL access, which is why the recovery test used that case.

8. Limitations and errors

- A logical dump captures one moment only.
- The restore ran as postgres, so objects belong to postgres, the same as the original.
- The restore and the README's --clean restore gave no errors or warnings.

- The test databases grp_brewposdb_restore and grp_brewposdb_recovery are dropped after the evidence is captured.

9. Feedback for the owners

Backup and restore worked first time. Suggested additions: a roles-only dump, re-applying 05_create_app_role.sql after a restore, and a dedicated backup role.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own computer, and every result in this report comes from the saved output files.